Green University of Bangladesh
Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering

Semester: (Spring, Year:2026), B.Sc. in CSE (Day)

LAB REPORT NO: 2

Course Title: Database Lab

Course Code: CSE-210 Section: 242-D3

Lab Experiment Name: Database Query using SQL Filters

Student Details

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Lab Date :

Submission Date : 13-04-2026

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Lab Report Status

Marks: …… **Signature:.....**

Comments:..... **Date:.....**

1.Title of The Lab Experiment

Database Query using SQL Filters

2.Objectives/ Aim:

To create a database using SQL
To design tables using (Primary Key, Foreign Key, UNIQUE, CHECK)
To insert multiple records into tables
To perform filtering operations using SQL queries
To use aggregate functions

3.PROCEDURE / ANALYSIS / DESIGN

Design

MEMBER(MemberID, FullName, Email, Age, Phone)

BOOK(BookID, Title, ISBN, Price, Availability)

BORROW(BorrowID, MemberID, BookID, BorrowDate)

4.IMPLEMENTATION

SQL CODE:

```
CREATE DATABASE library_query_lab;
USE library_query_lab;

CREATE TABLE MEMBER (
    MemberID INT PRIMARY KEY
    AUTO_INCREMENT,
    FullName VARCHAR(50) NOT NULL,
    Email VARCHAR(50) NOT NULL
    UNIQUE,
    Age INT CHECK (Age >= 12),
    Phone VARCHAR(15) NOT NULL
    DEFAULT 'N/A'
);

CREATE TABLE BOOK (
    BookID INT PRIMARY KEY
```

```
    AUTO_INCREMENT,
    Title VARCHAR(50) NOT NULL,
    ISBN VARCHAR(20) NOT NULL
    UNIQUE,
    Price DECIMAL(8,2) CHECK (Price >= 0),
    Availability
    ENUM('AVAILABLE','BORROWED')
    DEFAULT 'AVAILABLE'
);

CREATE TABLE BORROW (
    BorrowID INT PRIMARY KEY
    AUTO_INCREMENT,
    MemberID INT,
    BookID INT,
    BorrowDate DATE NOT NULL,
    FOREIGN KEY (MemberID)
    REFERENCES MEMBER(MemberID),
    FOREIGN KEY (BookID) REFERENCES
    BOOK(BookID)
);

INSERT INTO MEMBER (FullName, Email,
    Age, Phone) VALUES
('Amin', 'a1@mail.com', 22, '0171'),
('Bashir', 'b1@mail.com', 19, '0182'),
('Arman', 'a2@mail.com', 25, '0193'),
('Noyon', 'n1@mail.com', 17, '0164'),
('Sajid', 's1@mail.com', 21, '0155'),
('Alam', 'a3@mail.com', 30, '0146'),
('Tanvir', 't1@mail.com', 15, DEFAULT),
('Jibon', 'j1@mail.com', 28, '0137');

INSERT INTO BOOK (Title, ISBN, Price,
    Availability) VALUES
('DS', 'B001', 1200, 'AVAILABLE'),
```

```
('DB', 'B002', 1500, 'BORROWED'),
('Python', 'B003', 800, 'AVAILABLE'),
('OS', 'B004', 2000, 'BORROWED'),
('Net', 'B005', 1800, 'AVAILABLE'),
('DataSci', 'B006', 2200, 'AVAILABLE'),
('ML', 'B007', 2500, 'BORROWED'),
('Web', 'B008', 600, 'AVAILABLE');

INSERT INTO BORROW (MemberID,
    BookID, BorrowDate) VALUES
(1,2,'2026-01-01'),
(2,4,'2026-01-05'),
(3,7,'2026-01-10'),
(4,2,'2026-01-12'),
(5,4,'2026-01-15'),
(6,7,'2026-01-18'),
(1,4,'2026-01-20'),
(2,2,'2026-01-22'),
(3,7,'2026-01-25'),
(8,1,'2026-01-28');

SELECT * FROM MEMBER;
SELECT * FROM BOOK;
SELECT * FROM BORROW;

SELECT DISTINCT Availability FROM
    BOOK;

SELECT * FROM MEMBER WHERE Age >=
    18;

SELECT * FROM BOOK WHERE Price
    BETWEEN 500 AND 2000;

SELECT * FROM BOOK WHERE
    Availability IN
```

```
('AVAILABLE','BORROWED');

SELECT * FROM MEMBER WHERE
FullName LIKE 'A%' OR FullName LIKE
'%n';

SELECT * FROM MEMBER WHERE Phone
<> 'N/A' AND Age > 20;

SELECT * FROM MEMBER WHERE Age
NOT BETWEEN 12 AND 17;

SELECT * FROM BOOK WHERE Title LIKE
'%Data%' OR Title LIKE '%DB%';

SELECT * FROM BOOK ORDER BY Price
DESC LIMIT 3;

SELECT * FROM BOOK ORDER BY Price
ASC LIMIT 2;

SELECT
 (SELECT COUNT(*) FROM MEMBER) AS
   Members,
 (SELECT COUNT(*) FROM BOOK) AS
   Books,
 (SELECT COUNT(*) FROM BORROW) AS
   Borrows;

SELECT
 AVG(Price) AS AvgPrice,
 MIN(Price) AS MinPrice,
 MAX(Price) AS MaxPrice
FROM BOOK;

SELECT Availability, COUNT(*)
FROM BOOK
```

```
GROUP BY Availability;

SELECT MemberID, COUNT(*)
FROM BORROW
  GROUP BY MemberID;
```

5.TEST RESULT / OUTPUT

Action Output

#	Time	Action	Message
1	19:18:43	CREATE DATABASE library_query_lab	1 row(s) affected
2	19:18:43	USE library_query_lab	0 row(s) affected
3	19:18:43	CREATE TABLE MEMBER (MemberID INT PRIMARY KEY AUTO_INCREMENT, FullName VARCHAR(50) NOT NULL, Email VARCH...	0 row(s) affected
4	19:18:43	CREATE TABLE BOOK (BookID INT PRIMARY KEY AUTO_INCREMENT, Title VARCHAR(50) NOT NULL, ISBN VARCHAR(20) NO...	0 row(s) affected
5	19:18:43	CREATE TABLE BORROW (BorrowID INT PRIMARY KEY AUTO_INCREMENT, MemberID INT, BookID INT, BorrowDate DATE ...	0 row(s) affected
6	19:18:43	INSERT INTO MEMBER (FullName, Email, Age, Phone) VALUES ('Amin', 'a1@mail.com', 22, '0171'), ('Bashir', 'b1@mail.com', 19, '0182'), ('Am...	8 row(s) affected Re
7	19:18:43	INSERT INTO BOOK (Title, ISBN, Price, Availability) VALUES ('DS', 'B001', 1200, 'AVAILABLE'), ('DB', 'B002', 1500, 'BORROWED'), ('Python', ...	8 row(s) affected Re
8	19:18:43	INSERT INTO BORROW (MemberID, BookID, BorrowDate) VALUES (1,2,'2026-01-01'), (2,4,'2026-01-05'), (3,7,'2026-01-10'), (4,2,'2026-01-12'...	10 row(s) affected R
9	19:18:43	SELECT * FROM MEMBER LIMIT 0, 1000	8 row(s) returned
10	19:18:43	SELECT * FROM BOOK LIMIT 0, 1000	8 row(s) returned
11	19:18:44	SELECT * FROM BORROW LIMIT 0, 1000	10 row(s) returned
12	19:18:44	SELECT DISTINCT Availability FROM BOOK LIMIT 0, 1000	2 row(s) returned
13	19:18:44	SELECT * FROM MEMBER WHERE Age >= 18 LIMIT 0, 1000	6 row(s) returned
14	19:18:44	SELECT * FROM BOOK WHERE Price BETWEEN 500 AND 2000 LIMIT 0, 1000	6 row(s) returned
15	19:18:44	SELECT * FROM BOOK WHERE Availability IN ('AVAILABLE','BORROWED') LIMIT 0, 1000	8 row(s) returned
16	19:18:44	SELECT * FROM MEMBER WHERE FullName LIKE 'A%' OR FullName LIKE '%n' LIMIT 0, 1000	5 row(s) returned
17	19:18:44	SELECT * FROM MEMBER WHERE Phone <> 'N/A' AND Age > 20 LIMIT 0, 1000	5 row(s) returned
18	19:18:44	SELECT * FROM MEMBER WHERE Age NOT BETWEEN 12 AND 17 LIMIT 0, 1000	6 row(s) returned
19	19:18:44	SELECT * FROM BOOK WHERE Title LIKE '%Data%' OR Title LIKE '%DB%' LIMIT 0, 1000	2 row(s) returned
20	19:18:44	SELECT * FROM BOOK ORDER BY Price DESC LIMIT 3	3 row(s) returned
21	19:18:44	SELECT * FROM BOOK ORDER BY Price ASC LIMIT 2	2 row(s) returned
22	19:18:44	SELECT (SELECT COUNT(*) FROM MEMBER) AS Members, (SELECT COUNT(*) FROM BOOK) AS Books, (SELECT COUNT(*) FROM BO...	1 row(s) returned
23	19:18:44	SELECT AVG(Price) AS AvgPrice, MIN(Price) AS MinPrice, MAX(Price) AS MaxPrice FROM BOOK LIMIT 0, 1000	1 row(s) returned
24	19:18:44	SELECT Availability, COUNT(*) FROM BOOK GROUP BY Availability LIMIT 0, 1000	2 row(s) returned
25	19:18:44	SELECT MemberID, COUNT(*) FROM BORROW GROUP BY MemberID LIMIT 0, 1000	7 row(s) returned

7.SUMMARY:

In this lab, we designed a library database using SQL. We created tables with proper constraints and inserted data into them. Various SQL queries were used to filter, sort, and analyze the data. This lab helped improve understanding of database design and query operations.